Decoupled-and-Coupled Networks: Self-Supervised Hyperspectral Image Super-Resolution With Subpixel Fusion

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***Abstract*— Enormous efforts have been recently made to super-resolve hyperspectral (HS) images with the aid of high spatial resolution multispectral (MS) images. Most prior works usually perform the fusion task by means of multifarious pixel-level priors. Yet, the intrinsic effects of a large distribution gap between HS–MS data due to differences in the spatial and spectral resolution are less investigated. The gap might be caused by unknown sensor-specific properties or highly mixed spectral information within one pixel (due to low spatial resolution). To this end, we propose a subpixel-level HS super-resolution (HS-SR) framework by devising a novel decoupled-and-coupled network (DC-Net), to progressively fuse HS–MS information from the pixel level to subpixel level and from the image level to feature level. As the name suggests, DC-Net first decouples the input into common (or cross-sensor) and sensor-specific components to eliminate the gap between HS–MS images before further fusion and then thoroughly blends them by a model-guided coupled spectral unmixing (CSU) net. More significantly, we append a self-supervised learning module behind the CSU net by guaranteeing material consistency to enhance the detailed appearance of the restored HS product. Extensive experimental results show the superiority of our method both visually and quantitatively and achieve a significant improvement in comparison with the state of the art (SOTA).**

***Index Terms*— Data fusion, deep learning (DL), hyperspectral (HS) image, self-supervised, spectral unmixing, super-resolution.**

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I. Introduction

RECENT hyperspectral (HS) images are characterized by abundant and detailed spectral information, which enables the recognition of materials at a more accurate level compared with RGB or multispectral (MS) images. In recent years, HS imaging has been garnering increasing attention in a wide range of applications related to computer vision, such as image classification [1], [2], [3], image super-resolution [4], [5], [6], object detection, and tracking [7], [8], [9], [10], to name a few.

HS images are capable of capturing more subtle discrepancies between different objects, which, to a great extent, is benefited from its high spectral resolution. As a trade-off, HS imaging systems are usually designed to acquire the data at a high spatial sampling distance (i.e., sacrificing spatial resolution), thereby limiting the range of potential applications in practice [11]. Fortunately, MS imaging systems, with a broader spectral bandwidth, can provide finer spatial information. Therefore, fusing the low-resolution HS (LrHS) and high-resolution MS (HrMS) image pair is an intuitive and feasible solution to generate the high-resolution HS (HrHS) product, also known as MS-guided HS super-resolution (HS-SR). Fig. 1(a) illustrates the observation models for HrMS and LrHS images degraded from the HrHS image.

Over the last decade, a variety of model-driven HS-SR algorithms have been successfully developed [12]. These approaches model the underlying relationships between HS–MS data by various handcrafted priors and the knowledge of relevant sensor characteristics, e.g., spectral response function (SRF) and point spread function (PSF) of different camera systems. Very recently, the great success of convolutional neural networks (CNNs) has also made significant progress on the HS-SR task [13], [14], [15], [16], [17]. Owing to the powerful representation ability, CNNs-based models can excavate the intrinsic properties that lie in the HS (or MS) image more effectively and blend HS and MS images in a more compact way. Nevertheless, those existing, either model-based or data-driven (e.g., CNNs), fusion approaches rarely consider underlying problems that impact the HS-SR performance to a great extent, e.g., different sensor-specific properties leading to the weak affinity between original HS and MS data, highly mixed pixels widely existed in both HS and MS images [see Fig. 1(a)]. Mixed pixels inevitably degrade the performance of HR-SR due to the inherent ambiguity in the types and

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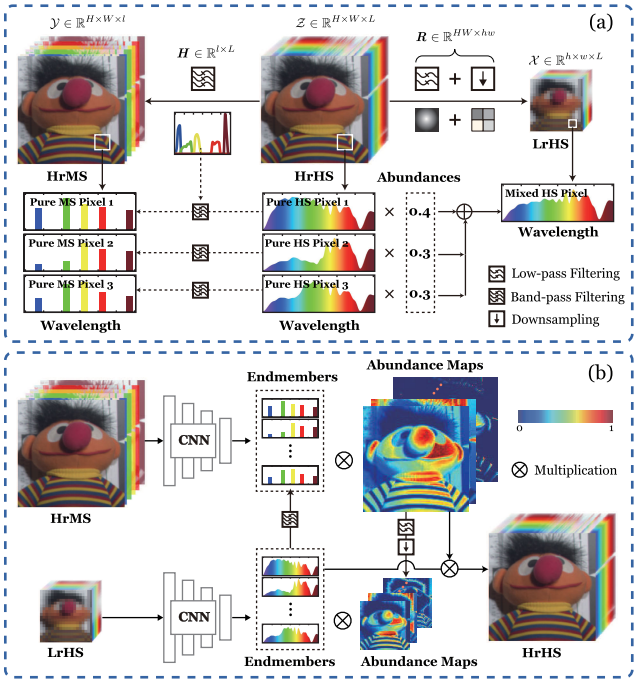


Fig. 1. (a) Observation models for HrMS and LrHS images and illustrates potential challenges (highly mixed spectral signatures) in HS-SR. (b) Solution by CSU to decompose mixed pixels to pure and informative subpixels, yielding a subpixel-level HS-SR approach.

quantities of materials within them. To address this challenge, a multitude of approaches have been developed for spectral unmixing and subpixel mapping, as demonstrated by previous works, such as [18], [19], [20], and [21]. These methods have proven to be effective and practical solutions for mitigating the impact of mixed pixels. They excel at decomposing mixed pixels into a distinct set of pure endmembers and their corresponding abundances, enabling the recovery of finer details from the subpixels.

To this end, we propose a decoupled-and-coupled network (DC-Net) for the HS-SR task. DC-Net is a novel progressive fusion framework from the pixel- to subpixel-level fusion and from the image- to feature-level fusion. As the name suggests, we first design a decoupled subnetwork (D-Net) by means of generative adversarial networks (GANs) [22]. The D-Net disentangles the input HS and MS images to common (or cross-sensor) and sensor-specific components and recombines them in a new image space, where the sensor-specific information is preliminarily exchanged to enable a tight connection of HS–MS information, further providing a better starting point for the subsequent fusion process. Inspired by spectral unmixing [23], we then develop a model-driven coupled subnetwork (C-Net) on the output of the D-Net to address the mixed pixel problem for a more compact fusion of HS–MS images. Fig. 1(b) illustrates a subpixel-level solution for HS-SR by the coupled spectral unmixing (CSU) net. More significantly, a self-supervised learning module (S-Net) is deployed to the C-Net to further enhance detailed geometries and textures, yielding more realistic HrHS images. The main contributions of this article are threefold as follows.

1) Beyond the pixel-level fusion, we provide a new subpixel insight for the HS-SR task by developing a unified DC-Net architecture in an end-to-end fashion.
2) DC-Net is a progressive fusion framework, which not only can eliminate the gap between HS–MS images for better pixel-level information exchange (from the image perspective via D-Net) but also can alleviate the effects of highly mixed spectral pixels (due to low spatial resolution) for a more compact subpixel-level fusion (from the feature perspective via C-Net).
3) A plug-and-play self-supervised module, i.e., S-Net, is devised in C-Net by aligning the abundance maps obtained by unmixing HS and MS images, respectively, to maintain semantic consistency between abundance maps of HS–MS images, further improving the quality of the HrHS product.

The rest of this article is organized as follows. Section II introduces the related works from the perspectives of traditional models and deep models in detail. We then elaborate on the proposed DC-Net and detail each newly developed module. Experiential results are conducted in comparison with currently existing state-of-the-art (SOTA) HS-SR approaches in Section IV. Finally, Section V draws a conclusion with a possible future outlook.

II. Related Work

A. Traditional Models

HS pansharpening is a heuristic way to perform the HS–MS fusion [24], which has been widely applied in the HS image super-resolution task. Component substitution (CS) and multiresolution analysis (MRA) are the two main types of pansharpening techniques. The former aims to inject detailed information about MS images into the LrHS image, thereby generating the HrHS product. The latter is to pansharpen the HS image by linearly combining MS bands to synthesize an HrHS band using regression techniques.

One representative technique that fuses the HS–MS images is the subspace-based approach, which is usually designed to enhance the spatial resolution of HS images by means of matrix factorization [11], [25], [26], [27], [28] or Bayesian estimation [29], [30]. Kawakami et al. [11] first decomposed the LrHS image into a spectral dictionary and corresponding sparse representations via matrix factorization, and the HrHS image can be then formed by using shareable sparse coefficients estimated from the RGB observation with a sampled spectral dictionary. Akhtar et al. [26] jointly considered different physical properties of materials in the scene, e.g., sparsity, non-negativity, and spatial structure, to improve the quality of the HrHS image. Dong et al. [28] structured the sparse coding model with the application to HS-SR. Yokoya's algorithm [25] is designed to perform the HS–MS fusion via an effective couple matrix factorization approach, which can be well explained from the spectral unmixing perspective. Similarly, Lanaras et al. [27] further extended [25] by jointly unmixing the input HS and MS data and fusing them in the latent subspace. In [29], the spectral dictionary is learned by using a parameter-free Bayesian model. Using the learned dictionary, the HrHS image can be reconstructed with potentially sparse assumptions. Furthermore, the same investigators used the Gaussian process as the priors on the basis of the

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Bayesian model to super-resolve the HS image [30]. Beyond matrix factorization, Zhang et al. [5] and Dian et al. [31] have attempted to model the HS-SR issue in the tensor form to achieve better structural representations.

B. DL-Based Models

These traditional methods have shown competitive performance in the HS–MS fusion task, yet they, to a great extent, rely on cross-sensor calibration and strong handcrafted priors. Inspired by the recent success of deep learning (DL) techniques, CNNs-based approaches have been garnering growing attention in the HS-SR task [13], [14], [15], [16], [17], [32]. Dian et al. [13] proposed a two-stage HS image pansharening framework by using CNNs-based prior training to refine the initialized fusion results obtained by traditional optimization methods. Qu et al. [14] trained an end-to-end unsupervised Dirichlet network to restore the HrHS image. Xie et al. [15] sought to open the "black box" and presented an interpretable deep supervised model for the MS/HS fusion task. Yao et al. [16] attempted to explain an unsupervised deep super-resolution network from the spectral unmixing perspective by introducing cycle consistency and cross-attention mechanism. Zhu et al. [32] proposed a supervised deep residual network in a progressive way for the HS image super-resolution. Furthermore, the field of RS has witnessed numerous advancements in multimodal image fusion and analysis techniques [33], [34], [35], [36]. These sophisticated approaches have found widespread application in various practical scenarios, including pansharpening, classification, unmixing, geospatial object detection, and more.

III. Decoupled-and-Coupled Networks

A. Overview

A large distribution gap between HS–MS images possibly caused by unknown sensor-specific information (e.g., HS–MS) and highly mixed spectral pixels has been shown to be a challenging and potential problem in the HS-SR task [12], [25], [27]. Focusing on it, we provide a point-to-point solution by designing a novel network architecture, i.e., DC-Net, which consists of two subnetworks. D-Net aims to decompose the original HS–MS data into the common content (or structure) and sensor-specific features and implement crossover recombination on the sensor-specific information to improve the compatibility between HS–MS images for better subsequent fusion. With the input of recombined HS–MS images, C-Net performs the subpixel-level fusion by decomposing the mixed pixels into fractional coefficients (termed abundances) and pure spectral signatures (termed endmembers), yielding an interpretable CSU-guided HS-SR network. It should be noted that DC-Net (i.e., D-Net and C-Net) can be regarded as a novel progressive fusion framework from pixel level to subpixel level and from the image level to feature level. Furthermore, a simple but effective self-supervised subnetwork (S-Net) in C-Net is devised to improve the quality of the HrHS product by correcting the material correspondence between HS–MS images. Fig. 2 illustrates the architecture overview of the proposed DC-Net.

B. Problem Formulation

Let $\mathcal{X} \in \mathbb{R}^{h\times w\times L}$, $\mathcal{Y} \in \mathbb{R}^{H\times W\times l}$, and $\mathcal{Z} \in \mathbb{R}^{H\times W\times L}$ be LrHS with hw pixels by L bands, HrMS with HW pixels by l bands, and HrHS images with HW pixels by L bands, respectively ($h < H$, $w < W$, and $l < L$); then $\boldsymbol{X} \in \mathbb{R}^{L\times hw}$, $\boldsymbol{Y} \in \mathbb{R}^{l\times HW}$, and $\boldsymbol{Z} \in \mathbb{R}^{L\times HW}$ are defined as the unfolded matrices. The observation models for LrHS and HrMS images from the HrHS image can be written as follows:

$$\begin{aligned} \boldsymbol{X} &= \boldsymbol{ZR} + \boldsymbol{N}_x \\ \boldsymbol{Y} &= \boldsymbol{HZ} + \boldsymbol{N}_y \end{aligned} \tag{1}$$

where $\boldsymbol{H} \in \mathbb{R}^{l\times L}$ and $\boldsymbol{R} \in \mathbb{R}^{HW\times hw}$ denote the SRF and PSF to degrade the spectral and spatial resolutions of $\boldsymbol{Z}$, respectively, and $\boldsymbol{N}_x$ and $\boldsymbol{N}_y$ are the observed noises.

Ideally, the observations in (1) can be further unfolded by using the spectral mixing model as follows:

$$\begin{aligned} \boldsymbol{X} &= \boldsymbol{SAR} + \boldsymbol{N}_x = \boldsymbol{S}\tilde{\boldsymbol{A}} + \boldsymbol{N}_x \\ \boldsymbol{Y} &= \boldsymbol{HSA} + \boldsymbol{N}_y = \tilde{\boldsymbol{S}}\boldsymbol{A} + \boldsymbol{N}_y \end{aligned} \tag{2}$$

where $\boldsymbol{S} \geq \boldsymbol{0}$ and $\boldsymbol{A} \geq \boldsymbol{0}$ denote a collection of pure spectral signatures (i.e., endmembers) and corresponding fractional coefficients (i.e., abundances) that meet the sum-to-one constraint (i.e., $\boldsymbol{1}^\top \boldsymbol{A} = \boldsymbol{1}^\top$), respectively. The two constraints offer valuable physical insights regarding the endmembers and abundances, enhancing the interpretability of our observation model and ensuring its suitability for real-world scenarios. $\tilde{\boldsymbol{S}}$ and $\tilde{\boldsymbol{A}}$ can be explained as the degraded $\boldsymbol{S}$ and $\boldsymbol{A}$ in spectral and spatial domains with the use of SRF and PSF. Thus, estimating $\boldsymbol{Z}$ is equivalent to finding $\boldsymbol{S}$ and $\boldsymbol{A}$, i.e., $\boldsymbol{Z} = \boldsymbol{SA}$, by solving problem (2).

Model (2) has been proven to be effective for the problem of mixed pixels in the HS–MS fusion [12], [14], [16], [25], [27], [31]. However, it seldom considers the high coupling between pixels in the same data source. For this reason, we learn the mappings (f_x and f_y) for transforming $\boldsymbol{X}$ and $\boldsymbol{Y}$ to a latent image space (e.g., $\bar{\boldsymbol{X}}$ and $\bar{\boldsymbol{Y}}$), where the pixel information can be easily separated between the same data source and better fused across different data sources. Therefore, the resulting model is

$$\begin{aligned} \bar{\boldsymbol{X}} &= f_x(\boldsymbol{X}) = f_x\big(\boldsymbol{S}\tilde{\boldsymbol{A}} + \boldsymbol{N}_x\big) \\ \bar{\boldsymbol{Y}} &= f_y(\boldsymbol{Y}) = f_y\big(\tilde{\boldsymbol{S}}\boldsymbol{A} + \boldsymbol{N}_y\big). \end{aligned} \tag{3}$$

In the following, we will build the DC-Net architecture step-by-step to perform the HS–MS fusion in model (3).

C. Decoupled Network (D-Net)

Owing to the powerful image-to-image transformation ability of GANs [37], [38], [39], we develop the D-Net to decouple the input images and recombine them (see Fig. 2), making it possible for the recombined images to be fused more sufficiently. D-Net embeds the HS–MS images onto a common content space $\mathcal{C}$ and sensor-specific spaces, e.g., $\mathcal{S}_\mathcal{X}$ and $\mathcal{S}_\mathcal{Y}$. This process can be performed by cross-sensor (or domain-common) encoders $\{E^{\mathcal{C}}_\mathcal{X}, E^{\mathcal{C}}_\mathcal{Y}\}$, sensor-specific encoders $\{E^{\mathcal{S}}_\mathcal{X}, E^{\mathcal{S}}_\mathcal{Y}\}$, generators $\{G_\mathcal{X}, G_\mathcal{Y}\}$, and

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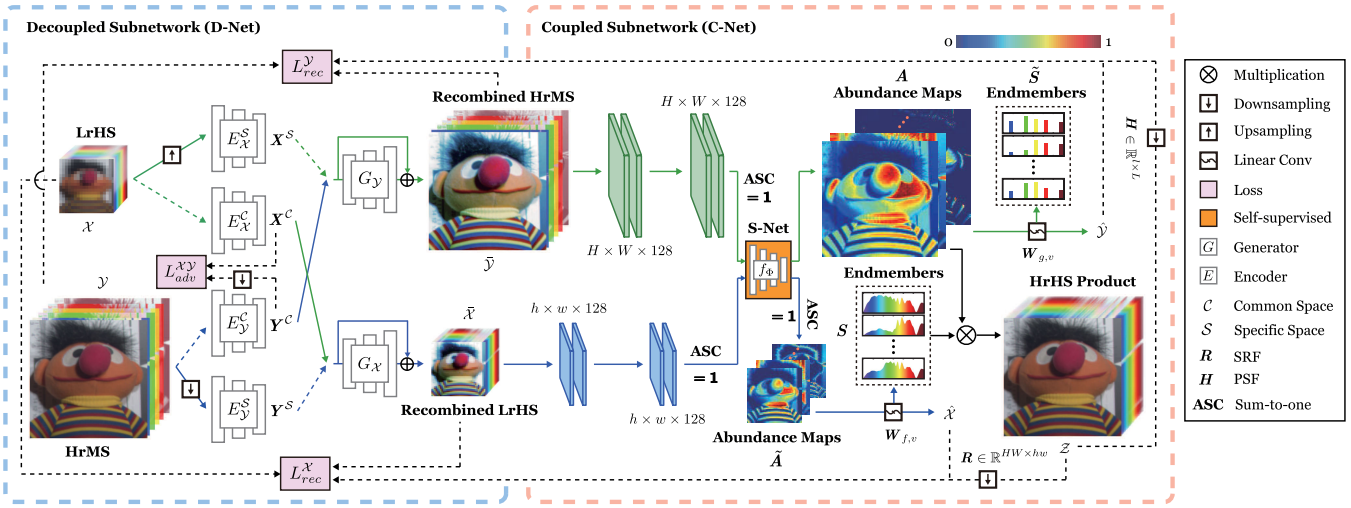


Fig. 2. Illustrative overview of the proposed DC-Net. It consists of two subnetworks: D-Net and C-Net. With the adversarial loss $L_{\text{adv}}^{\mathcal{XY}}$ between the common contents ($\bar{\boldsymbol{X}}$ and $\bar{\boldsymbol{Y}}$) of $\boldsymbol{X}$ and $\boldsymbol{Y}$, the reconstruction loss L_{rec}, and the self-supervised module (i.e., S-Net), we can learn an end-to-end HS-SR network more effectively without any prior training and better enhance detailed appearances.

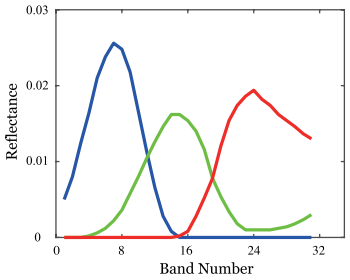


Fig. 3. Example for SRF (i.e., $\boldsymbol{H}$) in RGB bands.

a domain-common discriminator $D_{\text{adv}}^{\mathcal{C}}$. Given the input data ($\boldsymbol{X}$ and $\boldsymbol{Y}$), we then have the following transformations:

$$\begin{aligned}\left\{\boldsymbol{X}^{\mathcal{C}}, \boldsymbol{X}^{\mathcal{S}}\right\} &= \left\{E_{\mathcal{X}}^{\mathcal{C}}(\boldsymbol{X}), E_{\mathcal{X}}^{\mathcal{S}}(\boldsymbol{X}_{\uparrow})\right\}\\ \left\{\boldsymbol{Y}^{\mathcal{C}}, \boldsymbol{Y}^{\mathcal{S}}\right\} &= \left\{E_{\mathcal{Y}}^{\mathcal{C}}(\boldsymbol{Y}), E_{\mathcal{Y}}^{\mathcal{S}}(\boldsymbol{Y}_{\downarrow})\right\}\end{aligned} \tag{4}$$

where $\boldsymbol{X}^{\mathcal{C}} \in \mathcal{C}$ and $\boldsymbol{Y}^{\mathcal{C}} \in \mathcal{C}$ are the common information between $\boldsymbol{X}$ and $\boldsymbol{Y}$, and $\boldsymbol{X}^{\mathcal{S}} \in \mathcal{S}_{\mathcal{X}}$ and $\boldsymbol{Y}^{\mathcal{S}} \in \mathcal{S}_{\mathcal{Y}}$ are the domain-specific information.

By interactively recombining the common and sensor-specific information, we can further obtain transformed images, i.e., $\bar{\boldsymbol{X}}$ and $\bar{\boldsymbol{Y}}$, via generators ($\{G_{\mathcal{X}}, G_{\mathcal{Y}}\}$). This process can be formulated by $\bar{\boldsymbol{X}} = G_{\mathcal{X}}(\boldsymbol{X}^{\mathcal{C}}, \boldsymbol{Y}^{\mathcal{S}})$ and $\bar{\boldsymbol{Y}} = G_{\mathcal{Y}}(\boldsymbol{Y}^{\mathcal{C}}, \boldsymbol{X}^{\mathcal{S}})$. Such recombined images provide higher affinity between different sources (e.g., HS–MS), thereby yielding a greater potential for follow-up fusion.

D. Coupled Network (C-Net)

Linking to the D-Net, C-Net unmixes the recombined HS and MS images into endmembers ($\boldsymbol{S}$ and $\tilde{\boldsymbol{S}}$) and abundances ($\tilde{\boldsymbol{A}}$ and $\boldsymbol{A}$). As shown in Fig. 2, C-Net is a coupled convolutional autoencoder, which consists of two encoders (f_u and g_u) and two decoders (f_v and g_v) for $\bar{\boldsymbol{X}}$ and $\bar{\boldsymbol{Y}}$, respectively. This can be formulated accordingly as follows:

$$\begin{aligned}\hat{\boldsymbol{X}} &= \boldsymbol{S}\tilde{\boldsymbol{A}} = f\left(\bar{\boldsymbol{X}}\right) = f_v\left(f_u\left(\bar{\boldsymbol{X}}; \boldsymbol{W}_{f,u}\right); \boldsymbol{W}_{f,v}\right)\\ \hat{\boldsymbol{Y}} &= \tilde{\boldsymbol{S}}\boldsymbol{A} = g\left(\bar{\boldsymbol{Y}}\right) = g_v\left(g_u\left(\bar{\boldsymbol{Y}}; \boldsymbol{W}_{g,u}\right); \boldsymbol{W}_{g,v}\right).\end{aligned} \tag{5}$$

$\boldsymbol{W}$ is the weights of C-Net, where $\boldsymbol{W}_{f,v}$ and $\boldsymbol{W}_{g,v}$ are specified to be a 1×1 linear convolutional kernel, which can be well explained as endmembers, i.e., $\boldsymbol{S}$ and $\tilde{\boldsymbol{S}}$. Note that we adopt the clamp function to meet the abundance non-negative constraint (ANC), while the abundance sum-to-one constraint (ASC), i.e., $\boldsymbol{1}^{\top}\boldsymbol{A} = \boldsymbol{1}^{\top}$, can be guaranteed in the form of regularization. Using (5), the to-be-estimated HrHS image $\hat{\boldsymbol{Z}}$ can be obtained by $\hat{\boldsymbol{Z}} = \boldsymbol{S}\boldsymbol{A} = f_v(g_u(\bar{\boldsymbol{Y}}; \boldsymbol{W}_{g,u}); \boldsymbol{W}_{f,v})$.

Cycle Consistency: Coupled factors, i.e., SRF ($\boldsymbol{H}$) and PSF ($\boldsymbol{R}$), play a dominant role in the performance improvement of C-Net. Fig. 3 shows an SRF example with respect to RGB bands. Unlike previous methods that assume the two functions to be known, we attempt to automatically learn them by designing a cycle consistency mechanism in networks, enabling it to be more applicable in reality. For simplicity, the cycle can be written as $(\boldsymbol{X}, \boldsymbol{Y}) \xrightarrow{\text{D-Net}} (\bar{\boldsymbol{X}}, \bar{\boldsymbol{Y}}) \xrightarrow{\text{C-Net}} \hat{\boldsymbol{Z}} \xrightarrow{(\boldsymbol{H},\boldsymbol{R})} (\boldsymbol{X}, \boldsymbol{Y})$. More specifically, given the spectrum of the ith pixel in $\boldsymbol{Z}$ (denoted as $\boldsymbol{z}_i$) and the corresponding spectrum of the ith pixel and the jth channel in $\boldsymbol{Y}$ (denoted as $y_{i,j}$), the degradation process can be then expressed as follows:

$$y_{i,j} = \int_{\phi} \frac{\boldsymbol{h}_j(\eta)}{N_h} \boldsymbol{z}_i(\eta) d\eta \tag{6}$$

where ϕ denotes the support set with respect to the wavelength η, $\int \boldsymbol{h}_j(\eta) d\eta$ is the continuous representation of $\boldsymbol{H}$, and N_h is a constant for scaled normalization. In our case, $\boldsymbol{H}$ can be approximated using a set of $1 \times 1 \times L$ convolutional kernels, e.g., $\boldsymbol{w}_j$, in a discretized way, i.e.,

$$y_{i,j} = \boldsymbol{z}_j \otimes \boldsymbol{w}_j = \sum_{\phi} \frac{\boldsymbol{w}_j(\eta)}{N_w} \boldsymbol{z}_i(\eta). \tag{7}$$

Similarly, the spatial downsampling process performed by the PSF ($\boldsymbol{R}$) can also be simulated by means of convolutional operators with the same scaling kernel and stride sizes.

As a result, the cycle consistency for both spatial and spectral degradation can be well organized in the C-Net as follows:

$$\boldsymbol{X} = \hat{\boldsymbol{Z}}\boldsymbol{R}, \quad \boldsymbol{Y} = \boldsymbol{H}\hat{\boldsymbol{Z}} \tag{8}$$

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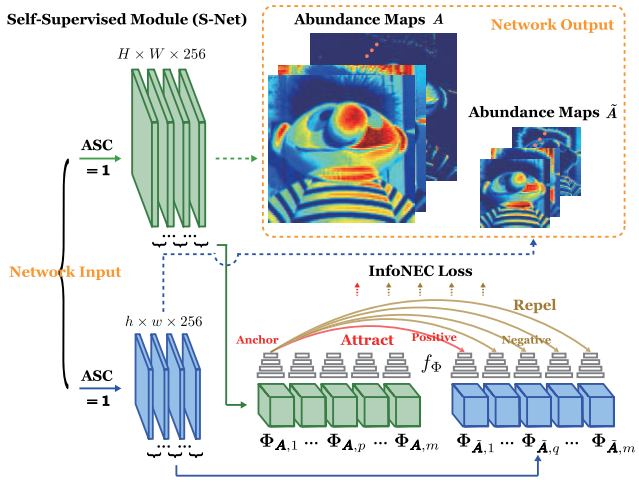


Fig. 4. Illustration of S-Net module.

which is a good fit for the observation models in (1) and helps us form a closed cycle chain.

E. Self-Supervised Subnetwork (S-Net)

In C-Net, the spectral unmixing process for HS and MS images is relatively independent. This might lead to chaotic semantic correspondences on abundance maps (or endmembers) of HS–MS images, e.g., between $\boldsymbol{A}$ and $\tilde{\boldsymbol{A}}$ or $\tilde{\boldsymbol{S}}$ and $\boldsymbol{S}$. To this end, we design an effective self-supervised learning module (S-Net) in C-Net to obtain more plausible abundance maps and endmembers, further yielding a more accurate HS reconstruction result. Fig. 4 illustrates the S-Net.

S-Net aims at correcting the correspondence of abundance maps of HS–MS images by learning high-level semantic alignment or consistency (in both order and category of materials), enabling a one-to-one match between $\boldsymbol{A}$ and $\tilde{\boldsymbol{A}}$. We call it as material or semantic consistency. The abundance maps of the same materials should be attracted to each other (as positive samples); otherwise, they are repelled (as negative samples). Considering the computational cost (since hundreds of abundance maps and spectral bundles usually need to be considered), we employ the same grouping strategy on $\boldsymbol{A}$ and $\tilde{\boldsymbol{A}}$ to make a group-to-group match. $\boldsymbol{A}$ and $\tilde{\boldsymbol{A}}$ can be grouped as follows:

$$\begin{aligned}\boldsymbol{A} &= [\underbrace{\boldsymbol{A}_1, \ldots}_{\boldsymbol{\Phi}_{A,1}}, \ldots, \underbrace{\boldsymbol{A}_k, \ldots}_{\boldsymbol{\Phi}_{A,p}}, \ldots, \underbrace{\ldots, \boldsymbol{A}_n}_{\boldsymbol{\Phi}_{A,m}}] \\ \tilde{\boldsymbol{A}} &= [\underbrace{\tilde{\boldsymbol{A}}_1, \ldots}_{\boldsymbol{\Phi}_{\tilde{A},1}}, \ldots, \underbrace{\tilde{\boldsymbol{A}}_k, \ldots}_{\boldsymbol{\Phi}_{\tilde{A},q}}, \ldots, \underbrace{\ldots, \tilde{\boldsymbol{A}}_n}_{\boldsymbol{\Phi}_{\tilde{A},m}}]\end{aligned} \tag{9}$$

where $\boldsymbol{\Phi}_{A,p}$ and $\boldsymbol{\Phi}_{\tilde{A},q}$ denote the grouping sets with respect to $\boldsymbol{A}$ and $\tilde{\boldsymbol{A}}$ for $\forall p, q \in \{1, \ldots, m\}$, respectively. We experimentally set the grouping parameter m to be 8. Furthermore, the representation vectors of $\boldsymbol{\Phi}_{A,p}$ and $\boldsymbol{\Phi}_{\tilde{A},q}$, denoted as z, can be obtained via a two-stream encoder or CNN, e.g., $f_{\boldsymbol{\Phi}_A}$ and $f_{\boldsymbol{\Phi}_{\tilde{A}}}$, which both consists of two blocks, i.e., [conv-relu-pool-fc]. The main difference lies in the pooling size and stride for $f_{\boldsymbol{\Phi}_A}$ larger than those for $f_{\boldsymbol{\Phi}_{\tilde{A}}}$ to guarantee the same size for final output features. We then have $z_{A,p} = f_{\boldsymbol{\Phi}_A}(\boldsymbol{\Phi}_{A,p})$ and $z_{\tilde{A},q} = f_{\boldsymbol{\Phi}_{\tilde{A}}}(\boldsymbol{\Phi}_{\tilde{A},q})$. In our S-Net, only when $p = q$, $z_{\tilde{A},q}$ is the positive sample of $z_{A,p}$; otherwise, $(p \neq q)$, and they are the negative sample pair.

TABLE I

QUANTITATIVE AVERAGE PERFORMANCE OF SOTA SUPER-RESOLUTION METHODS OVER 16 TESTING IMAGES IN TERMS OF FIVE DIFFERENT PQIS ON THE CAVE DATASET

Method	Ref.	Metric				
		PSNR	SAM	ERGAS	SSIM	UQI
BSR	CVPR'15 [29]	31.08	13.52	0.72	0.901	0.894
CSU	ICCV'15 [27]	29.69	13.13	0.70	0.891	0.900
NSSR	TIP'16 [28]	33.93	10.87	0.61	0.925	0.914
NLSTF	CVPR'17 [31]	30.31	12.29	0.69	0.914	0.896
uSDN	CVPR'18 [14]	35.60	9.97	0.53	0.934	0.921
MHF-Net	CVPR'19 [15]	37.41	7.33	0.45	0.965	0.958
GDD	ECCV'20 [17]	37.06	6.71	0.40	0.971	0.960
CUCaNet	ECCV'20 [16]	37.29	7.03	0.46	0.960	0.951
DC-Net	–	38.12	6.48	0.39	0.973	0.977
DC-Net-S	–	**38.68**	**6.15**	**0.34**	**0.978**	**0.980**
Ideal Value	–	$\uparrow$	$\downarrow \mathbf{0}$	$\downarrow \mathbf{0}$	$\uparrow \mathbf{1}$	$\uparrow \mathbf{1}$

F. Network Training

1) Loss Functions: As shown in Fig. 2, there are several important loss functions that need to be carefully considered.

In D-Net, the adversarial loss between the common content of $\boldsymbol{X}$ and $\boldsymbol{Y}$ can be expressed as follows:

$$\begin{aligned}\mathcal{L}_{\text{adv}} = \mathbb{E}_x\left[\frac{1}{2}\log D^{\mathcal{C}}\left(\boldsymbol{X}^{\mathcal{C}}\right) + \frac{1}{2}\log\left(1 - D^{\mathcal{C}}\left(\boldsymbol{X}^{\mathcal{C}}\right)\right)\right] \\ + \mathbb{E}_y\left[\frac{1}{2}\log D^{\mathcal{C}}\left(\boldsymbol{Y}_{\downarrow}^{\mathcal{C}}\right) + \frac{1}{2}\log\left(1 - D^{\mathcal{C}}\left(\boldsymbol{Y}_{\downarrow}^{\mathcal{C}}\right)\right)\right]\end{aligned} \tag{10}$$

where $\boldsymbol{X}^{\mathcal{C}} = E_{\mathcal{X}}^{\mathcal{C}}(\boldsymbol{X})$ and $\boldsymbol{Y}^{\mathcal{C}} = E_{\mathcal{Y}}^{\mathcal{C}}(\boldsymbol{Y})$.

The reconstruction loss stemmed from coupled convolutional autoencoders and the cycle consistency can be computed by means of (5) and (8)

$$\begin{aligned}\mathcal{L}_{\text{rec}} = \left\|\hat{\boldsymbol{X}} - \boldsymbol{X}\right\|_1 + \left\|\hat{\boldsymbol{Y}} - \boldsymbol{Y}\right\|_1 + \left\|\hat{\boldsymbol{X}} - \boldsymbol{Z}_{\downarrow,\boldsymbol{R}}\right\|_1 \\ + \left\|\hat{\boldsymbol{Y}} - \boldsymbol{Z}_{\downarrow,\boldsymbol{H}}\right\|_1 + \left\|\bar{\boldsymbol{X}} - \boldsymbol{X}\right\|_1 + \left\|\bar{\boldsymbol{Y}} - \boldsymbol{Y}\right\|_1\end{aligned} \tag{11}$$

where the ℓ_1-norm measure is applied to enhance the detailed perception in image reconstruction [40], and $\hat{\boldsymbol{Z}}_{\downarrow,\boldsymbol{R}}$ and $\hat{\boldsymbol{Z}}_{\downarrow,\boldsymbol{H}}$ denote $\hat{\boldsymbol{Z}}\boldsymbol{R}$ and $\boldsymbol{H}\hat{\boldsymbol{Z}}$, respectively.

Moreover, the ASC is satisfied using the following loss:

$$\mathcal{L}_{\text{ASC}} = \left\|\mathbf{1}^\top - \mathbf{1}^\top \boldsymbol{A}\right\|_1 + \left\|\mathbf{1} - \mathbf{1}^\top \tilde{\boldsymbol{A}}\right\|_1. \tag{12}$$

In S-Net, the InfoNCE loss [41] is optimized to measure the distances between positive and negative samples

$$\mathcal{L}_{\text{self}} = -\mathbb{E}_{\boldsymbol{\Phi}}\left[\log \frac{\exp\left(f_{\boldsymbol{\Phi}}\left(\boldsymbol{\Phi}_{\boldsymbol{A},p}\right)^\top f_{\boldsymbol{\Phi}}\left(\boldsymbol{\Phi}_{\tilde{A},q}^{+}\right)\right)}{\sum_{q=1}^{m}\exp\left(f_{\boldsymbol{\Phi}}\left(\boldsymbol{\Phi}_{\boldsymbol{A},p}\right)^\top f_{\boldsymbol{\Phi}}\left(\boldsymbol{\Phi}_{\tilde{A},q}\right)\right)}\right] \tag{13}$$

where $\boldsymbol{\Phi}_{A,p}$ and $\boldsymbol{\Phi}_{\tilde{A},q}^{+}$ denote the anchor sample and the positive sample, respectively.

In summary, our DC-Net is trained by minimizing the following overall loss parameterized by the set $\{\alpha, \beta, \gamma\}$:

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \alpha\mathcal{L}_{\text{adv}} + \beta\mathcal{L}_{\text{ASC}} + \gamma\mathcal{L}_{\text{self}}. \tag{14}$$

The penalty parameters are established through empirical and experimental fine-tuning. In network training, different

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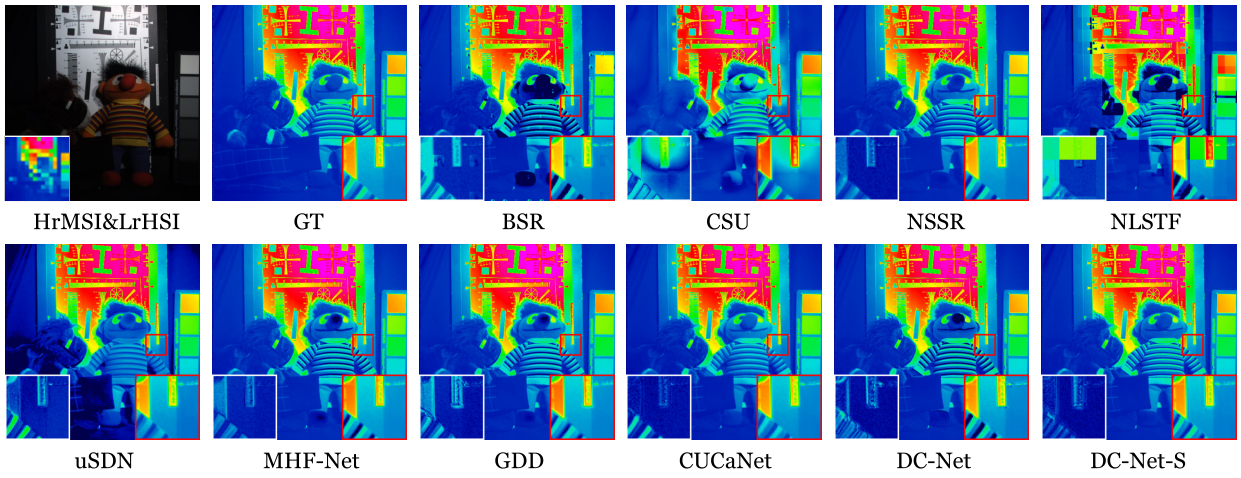


Fig. 5. Visualization of super-resolved HS images obtained by all compared methods (CAVE: chart and stuffed toy), where an ROI zoomed in four times (bottom right) and the difference image with GT (bottom left) are highlighted.

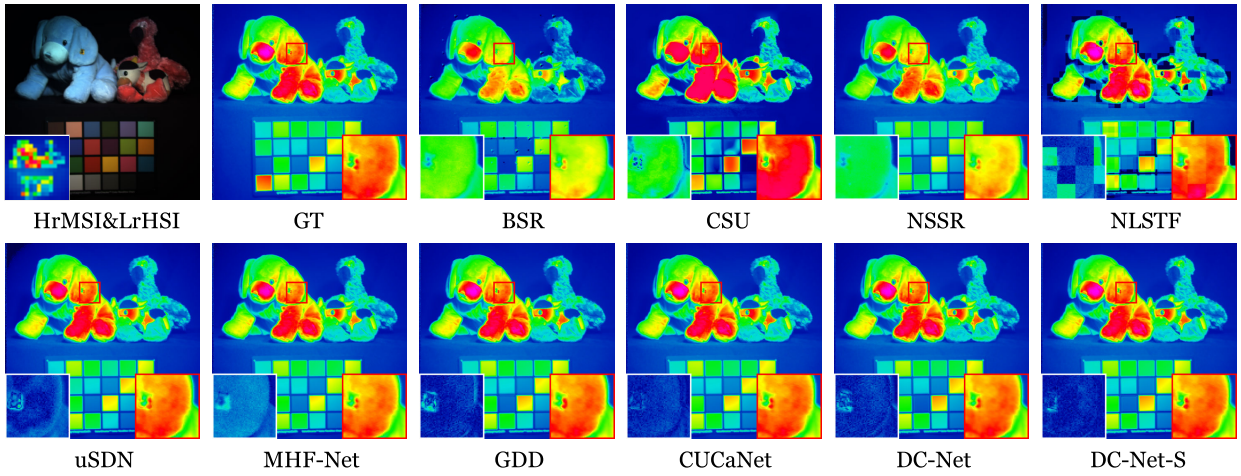


Fig. 6. Visualization of super-resolved HS images obtained by all compared methods (CAVE: stuffed toys), where an ROI zoomed in four times (bottom right) and the difference image with GT (bottom left) are highlighted.

TABLE II

QUANTITATIVE PERFORMANCE COMPARISON BETWEEN THE SAME SOTA SUPER-RESOLUTION METHODS IN TERMS OF FIVE DIFFERENT PQIS ON THE CHIKUSEI DATASET

Method	Ref.	Metric				
		PSNR	SAM	ERGAS	SSIM	UQI
BSR	CVPR'15 [29]	38.13	4.54	0.60	0.948	0.963
CSU	ICCV'15 [27]	36.96	5.20	0.73	0.898	0.937
NSSR	TIP'16 [28]	42.30	3.12	0.48	0.970	0.984
NLSTF	CVPR'17 [31]	37.04	6.11	0.70	0.901	0.925
uSDN	CVPR'18 [14]	40.02	2.96	0.41	0.964	0.981
MHF-Net	CVPR'19 [15]	42.43	3.01	0.44	0.977	0.982
GDD	ECCV'20 [17]	41.58	3.04	0.43	0.974	0.984
CUCaNet	ECCV'20 [16]	41.79	2.84	0.40	0.975	0.988
DC-Net	–	42.08	2.78	0.38	0.977	0.989
DC-Net-S	–	**42.56**	**2.55**	**0.34**	**0.979**	**0.992**
Ideal Value	–	↑	↓ **0**	↓ **0**	↑ **1**	↑ **1**

regularizations assume distinct roles, each making a substantial contribution to the overall process. Although their overall importance is generally on par, a more precise quantitative ranking of their significance is available in the following ablation analysis.

2) Implementation Details: Our networks are implemented on the PyTorch platform, and the Adam [42] optimizer is used to train the networks for 10 000 epochs with a batch size of 1. The base learning rate starts with 0.005, which can be gradually updated by a linear decay schedule [43]. We initialize all convolutional layers in our networks with Kaiming initialization [44]. Furthermore, the hyperparameters are experimentally determined using a grid search, and an early stopping strategy is considered in the network training when the validation loss fails to decrease.

IV. EXPERIMENTAL RESULTS

In this section, we first review the datasets and experimental settings. Next, we verify our networks with extensive ablation studies. Finally, the performance evaluation on simulated and real datasets is conducted to show the superiority of DC-Net in comparison with SOTA methods.

A. Dataset and Setup

Two public HS–MS datasets are used to evaluate the performance of DC-Net, i.e., CAVE dataset (indoor) [45] and Chikusei dataset (remote sensing) [12].

The CAVE dataset consists of 32 indoor HrHS images with the size of 512 × 512 × 31, covering the spectral wavelength

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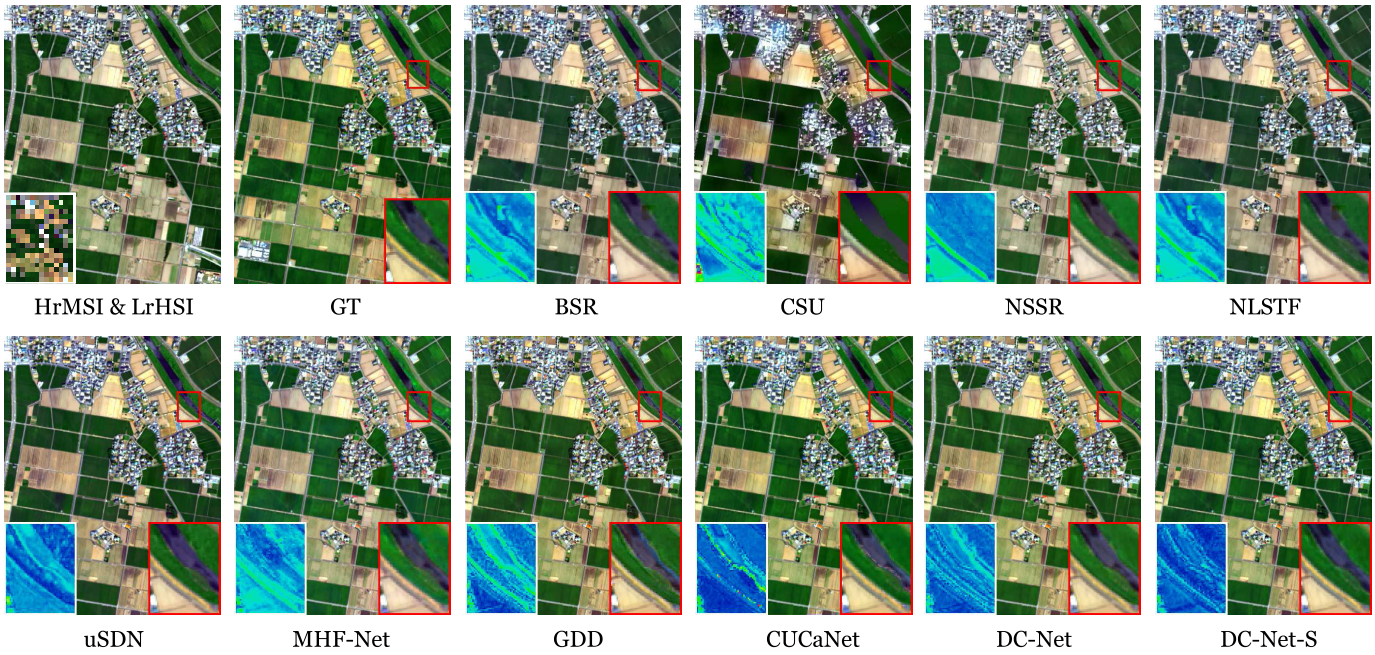


Fig. 7. Visualization (R: 63, G: 35, and B: 11) of super-resolved HS images obtained by all compared methods on the Chikusei dataset, where an ROI zoomed in four times and the difference image with GT are highlighted for more detailed observation.

from 400 to 700 nm. The Chikusei dataset acquired by the VNIR-C sensor comprises of 2517 × 2335 pixels and 128 spectral bands. We randomly crop the scene into 12 nonoverlapped subimages with the spatial size of 576 × 448. Moreover, we divide the two datasets into training, validation, and testing sets in the proportions of 8:8:16 and 4:4:4, respectively, where training and validation sets are used to determine optimal hyperparameters.

For the two datasets, the HrMS images are generated by degrading the HrHS images in the spectral domain with SRFs, while the LrHS images can also be simulated by spatially downsampling the HrHS images with the Wald et al.'s [46] protocol of a 32 sampling ratio. In addition, the Nikon D700 sensor and Landsat-8 MS sensor [47] are selected as the SRFs for the CAVE and Chikusei datasets, respectively.

Evaluation Metrics: We evaluate the fusion performance quantitatively in terms of five widely used picture quality indices (PQIs): peak signal-to-noise ratio (PSNR), spectral angle mapper (SAM) [48], erreur relative globale adimensionnelle de synthèse (ERGAS) [49], structure similarity (SSIM) [50], and universal image quality index (UIQI) [51].

B. Comparison With SOTAs

We evaluate the performance of DC-Net both quantitatively and qualitatively in comparison with SOTA models in the HS-SR task, including traditional SOTA methods: BSR [29],[1] CSU [27],[2] NSSR [28],[3] and NLSTF [31],[4] and DL-based SOTA methods: uSDN [14],[5] MHF-Net [15],[6] GDD [17],[7] and CUCaNet [16],[8] as well as the DC-Net without S-Net module. We maintain the same experimental configurations for all compared methods as in their original literature as much as possible.

1) Experiments on CAVE Data: Table I lists the average results over 16 testing images on the CAVE dataset to make a quantitative comparison between all compared SOTA methods and the DC-Net and DC-Net-S in terms of five PQIs.

Overall, BSR, as a pioneer work in HS-SR, reconstructs the high-quality HS image from the Bayesian statistic point of view and achieves competitive performance. Despite relatively lower PSNR and SSIM (see BSR), CSU can obtain better POIs in SAM, ERGAS, and UIQI, owing in large part to the use of the CSU mechanism. Beyond the CSU, NLSTF fuses HS and MS images in a coupled tensor form with nonlocal spatial information preservation, yielding better performance in all indices except the UIQI. By virtue of non-negativity and structured modeling, NSSR outperforms other traditional SOTA methods in HrHS image reconstruction. As an early stage DL-based method, the fusion performance of uSDN is moderately inferior to that of other DL-based competitors but is clearly higher than that of those non-DL algorithms. It is more noteworthy that the results obtained by unsupervised GDD and CUCaNet are basically comparable to those of the supervised MHF-Net (even better in certain indices). DC-Net behaves superiorly compared with the existing SOTA DL approaches, implying that the decoupled-and-coupled strategy creates the "one plus one greater than two" effect. Furthermore, S-Net module can learn high-level semantic consistency

[1]https://github.com/junjun-jiang/Hyperspectral-Image-Super-Resolution-Benchmark
[2]https:/github.com/lanha/SupResPALM
[3]http://see.xidian.edu.cn/faculty/wsdong
[4]https://github.com/renweidian/NLSTF
[5]https://github.com/aicip/uSDN
[6]https://github.com/XieQi2015/MHF-net
[7]https://github.com/tuezato/guided-deep-decoder
[8]https://github.com/danfenghong/ECCV2020_CUCaNet

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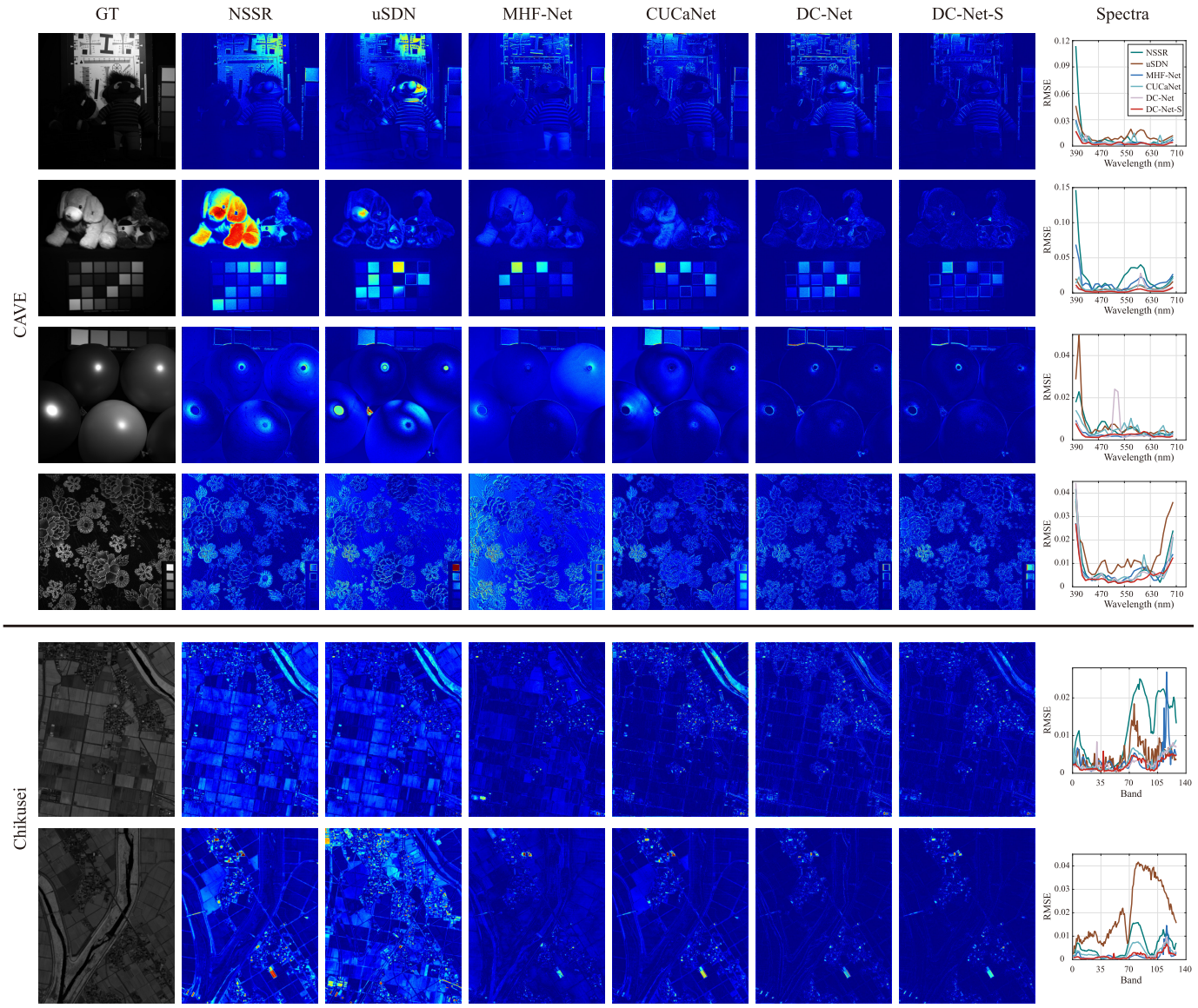


Fig. 8. Visual evaluation on CAVE and Chikusei datasets, respectively. The GT, the residual maps between GT maps, and super-solved HS results obtained by compared methods (e.g., NSSR, uSDN, MHF-Net, and CUCaNet) and our proposed DC-Net and DC-Net-S. The RMSE values of HS images for these methods along the spectra (covering the wavelength) are also given corresponding to the six studied scenes.

from abundance maps and embed interpretable and physically meaningful information into the DC-Net, enabling the best performance (i.e., DC-Net-S).

Figs. 5 and 6 visualize the super-resolved HS image in the 11th band (≈500 nm) for chart and stuffed toy. Furthermore, two regions of interest (ROIs) are selected to highlight the visually detailed differences of all compared methods. By and large, DL-based models outperform evidently compared with traditional methods, particularly in the coarser-grained structure of objects. The results of the proposed methods are superior to other unsupervised ones (e.g., uSDN, GDD, and CUCaNet), while DC-Net-S performs better than DC-Net in terms of detailed appearances. Despite similar visual results (see MHF-Net), our networks, either DC-Net or DC-Net-S, are capable of better recovering textural details and color cues that approximate ground truth (GT).

2) Experiments on Chikusei Dataset: Quantitative performance comparison on the Chikusei dataset is given in Table II, where there is a basically identical trend with that on the CAVE dataset (see Table I). Remote sensing images (due to the lower spatial resolution) are not as complicated as indoor images; all these methods, therefore, tend to super-resolve HS images with higher quality on the Chikusei dataset (see CAVE). Significantly, the performance of our DC-Net-S is still superior to other competitors with respect to all PQIs, demonstrating its superiority in the HS-SR task. However, the most noteworthy point lies in that MHF-Net obtains competitive results (fully comparable to DC-Net without S-Net module), since its accuracy relies greatly on supervised training on plenty of sample pairs. Fig. 7 gives a scene example and shows the composite HrHS images obtained by all competing methods with an ROI zoomed in four times for better observing visual differences.

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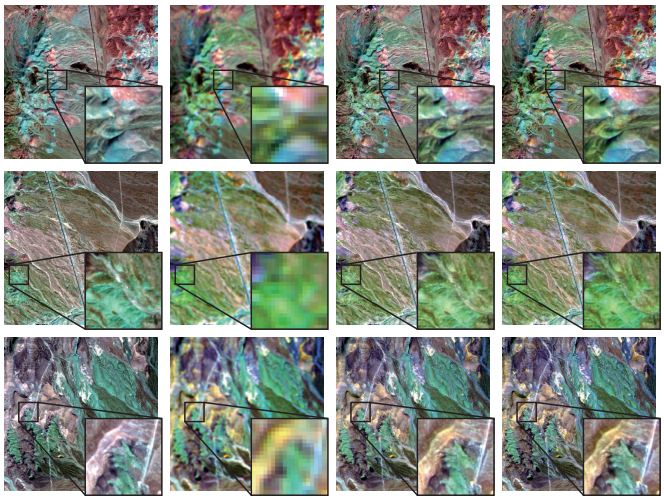


Fig. 9. Visualization of super-resolved HrHS fused by WV-3 HrMS (image courtesy Maxar) and Hyperion LrHS using our method (real data of Cuprite: three subregions).

TABLE III
ABLATION ANALYSIS OF THE PROPOSED DC-NET WITH A COMBINATION OF DIFFERENT MODULES ON THE CAVE DATASET

Method	Module					Metric	
	ANC	ASC	D-Net	C-Net	S-Net	PSNR	SAM
C-Net	✗	✗	✗	✓	✗	32.56	14.80
C-Net	✓	✗	✗	✓	✗	36.69	7.43
C-Net	✓	✓	✗	✓	✗	36.97	6.84
DC-Net	✓	✓	✓	✓	✗	38.36	4.77
DC-Net-S	✓	✓	✓	✓	✓	**39.15**	**4.51**
Ideal Value	–	–	–	–	–	↑	↓ **0**

3) Visual Quality Evaluation: To evaluate the visual quality of HS super-solved results more finely, we select several representative images from CAVE (four scenes) and Chikusei (two scenes) datasets, respectively, and show the corresponding residual maps between GT maps and super-solved HS results obtained by different compared methods, e.g., NSSR, uSDN, MHF-Net, and CUCaNet, and our proposed DC-Net and DC-Net-S, in Fig. 8. Furthermore, the RMSE values of HS images in each band are shown along spectra corresponding to the aforementioned six scenes. Intuitively, our methods perform better than other competitors from the perspective of residual maps, which demonstrates that our methods can provide finer spatial details. More importantly, the stability and generalization ability of our DC-Nets are higher than compared networks in different scenes, showing the effectiveness to a great extent. The bandwise RMES results obtained by our models are obviously similar and closer to the GT in the six investigated scenes. This, to some extent, indicates that the sequentially spectral information can be well recovered and retained by our approaches.

C. Evaluation on Real Data

To effectively assess the performance of our DC-Net, we curated a dataset comprising real, registered MS and HS scenes acquired from the WorldView-3 and Hyperion satellite missions, respectively. We focused on three specific subregions to facilitate a visual quality comparison of the composite HrHS products, as depicted in Fig. 9. Each subregion consists of HrMS and LrHS images, with the dimensions of 640 × 640 pixels and eight bands, maintaining a ground sampling distance (GSD) of 7.5 m, and 160 × 160 pixels with 167 spectral channels at a GSD of 30 m, respectively. Unfortunately, a real HrHS image is not available for reference.

In Fig. 9, it is evident that both DC-Net and DC-Net-S successfully restore HrHS results. While both methods produce clear results, DC-Net-S exhibits a higher degree of visual fidelity. In comparison with DC-Net, DC-Net-S achieves structural and textural appearances that closely match the HrMS image and, in some aspects, even surpass it. Furthermore, its color and brightness more closely align with those of the LrHS image. This highlights the superiority of the proposed methods in the HS-SR task. However, it is important to acknowledge that some spectral degradation is visually apparent. Therefore, there is room for improvement in terms of reconstruction performance.

TABLE IV
PARAMETER SIZE AND RUNNING TIME OF OUR DC-NET-S

Model	Size (MB)	Average Run Time (min.)		
		CAVE	Chikusei	Cuprite
DC-Net-S	3.43	101	312	156

D. Model Analysis

1) Ablation Studies: We investigate the performance gain by stepwise adding different modules (or subnetworks, i.e., D-Net and S-Net) in networks. We also study the importance of two physical constraints (ANC and ASC) related to the spectral unmixing model and their effects on the quality of HS-SR. To verify the effectiveness of the ablation analysis, we report the average results in terms of PSNR and SAM indices over eight images on the validation set of the CAVE dataset by maximizing the stepwise performance using optimal parameters, as listed in Table III.

In detail, C-Net without ANC and ASC yields relatively poor performance, which, to some extent, indicates the importance of the two critical physical constraints in the spectral unmixing-inspired HS-SR network. By turning on the ANC, the fusion results of C-Net increase dramatically (beyond four PNSR values and around halved SAM value). It is better still that, the joint use of ANC and ASC can further bring the performance improvement in both PSNR and SAM compared with ANC used alone in C-Net. As expected, DC-Net after plugging the D-Net performs observably better than C-Net at an increase of about 1.5 PSNR value and a decrease of just over two SAM values. More remarkably, the self-supervised module (S-Net) is capable of breaking through the existing "bottleneck" in DC-Net by enforcing the semantic consistency on abundance maps ($\boldsymbol{A}$ and $\tilde{\boldsymbol{A}}$).

2) Computational Analysis: We list the size of network parameters (i.e., DC-Net-S) and the inference time per image on a PC with one NVIDIA GeForce GTX 1080Ti GPU in Table IV. Our network parameters only have 3.43 MB, and

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since our method is unsupervised, the inference time is also desirable and acceptable in such a PC setting.

V. Conclusion

In this article, we propose a novel subpixel-level HS-SR framework, i.e., DC-Net, by fully considering the effects of affinity of the same data sources and exclusivity across HS–MS data and mixed pixels within one pixel. Inspired by spectral unmixing, DC-Net is capable of utilizing the intrinsic properties of HS–MS images effectively for the fusion task, which is more applicable to various real cases. We further optimize the network performance by devising an effective self-supervised module. Extensive experiments conducted on simulated and real data show the superiority of DC-Net in HS-SR over current SOTA fusion methods.

We found, however, that there exists spectral degradation in the HS-SR products of the real data. Consequently, in our future endeavors, we will prioritize the development of novel modules or the adoption of advanced network architectures aimed at preserving spectral sequentiality to the greatest extent possible while enhancing resolution.

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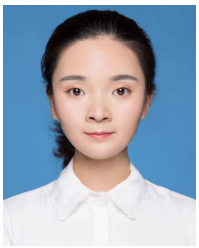
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